

NATHANIEL MCINTIRE

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EDUCATION

NORTH CAROLINA STATE UNIVERSITY

Expected Graduation: May 2027

Bachelor of Aerospace Engineering | Minor Physics | Minor Mathematics

- Relevant Coursework: Aerodynamics I & II, Aerospace Structures I & II, Automatic Controls, Space Flight, Thermodynamics, Numerical Analysis, FEA, Rocket Propulsion

BRUNSWICK COMMUNITY COLLEGE

May 2024

Associate of Engineering | Associate of Science | Cum Laude Honors

ENGINEERING PROJECTS & DESIGN EXPERIENCE

NASA RASC-AL LUNAR RESOURCE EXPLORATION ROVER - SENIOR DESIGN

- Contributing to a multidisciplinary engineering team for NASA RASC-AL Theme 3 to develop a prototype lunar south pole resource-acquisition rover.
- Establishing mission requirements and system concepts for terrain mapping, autonomous operation within permanently shadowed regions, and a reusable architecture for long-term lunar exploration.

NC STATE HIGH-POWERED ROCKETRY - IREC PAYLOAD SUBTEAM

- Contributing to NC State's inaugural International Rocket Engineering Competition (IREC) entry as a member of the payload subteam, supporting payload integration and system testing.
- Designed a compact KiCad PCB integrating two accelerometers, a barometer, and a magnetometer; supported ejection-system testing and carbon-fiber component fabrication.

AIRCRAFT WING AERODYNAMIC OPTIMIZATION

- Compared three airfoils in XFOIL across climb, cruise, and descent conditions and selected the NACA 23015 based on lift-to-drag performance, stability, and low-speed characteristics.
- Optimized a 50 m-span wing to AR 10 and a 0.5 taper ratio, calculating approximately 858 kN lift, 24.7 kN drag, and 138 kN-m wing-root torque at cruise.

CONVERGING-DIVERGING NOZZLE DESIGN AND FLOW ANALYSIS

- Designed and manufactured a converging-diverging nozzle in SolidWorks and developed an ANSYS CFD model to predict internal pressure and velocity behavior.
- Conducted wind-tunnel testing at NC State and processed experimental data in MATLAB to compare measured and computational flow characteristics.

COMPOSITE LAMINATE STRESS ANALYSIS

- Evaluated ply-level stresses in composite laminates under multiple loading conditions and transformed stresses between material and global coordinate systems.
- Investigated how ply orientation and loading direction affected the structural response of laminated composites.

IMPINGING-JET PRESSURE MEASUREMENT

- Measured the surface-pressure distribution produced by an impinging jet using binary pressure-sensitive paint.
- Developed a MATLAB image-processing workflow to convert experimental images into quantitative pressure-distribution data.

PENDULUM CONTROL-SYSTEM DESIGN

- Modeled a linearized pendulum system and designed a lead compensator using root-locus techniques to meet specified overshoot and settling-time requirements.
- Evaluated the compensated system response in MATLAB/Simulink and adjusted controller parameters to achieve the target transient behavior.

FLIGHT-DELAY PREDICTION MODELING

- Engineered predictive models for commercial aviation delays using Gini and entropy-based CART alongside random-forest algorithms.
- Utilized SMOTE and Tomek Links techniques to mitigate dataset class imbalances and conducted comparative performance assessments based on standardized metrics.

TECHNICAL SKILLS

MATLAB | Python | ANSYS | FEA | CFD | XFOIL | KiCAD | AutoCAD | SolidWorks | Numerical Analysis | Structural Analysis | Composite Materials | Aerodynamics | Rocket Propulsion | Orbital Mechanics | Machine Learning | PLC/Ladder Logic | Project Management |

PROFESSIONAL EXPERIENCE

BONGIORNO & SON ITALIAN SPECIALTIES

March 2025 - Present

Supervisor

Raleigh, NC

- Supervised a 10-person team and oversaw scheduling, inventory, cash handling, opening/closing procedures, and daily sales of \$3,000–\$6,500; reduced labor costs by 10% while increasing sales by 25% over a three-month period.

ACE HARDWARE

August 2024 - March 2025

Supervisor

Raleigh, NC

- Supervised 10 employees and coordinated inventory and deliveries, assembled approximately 20 mechanical products weekly, and operated a forklift for loads of up to 1200 lbs.

LOWES FOODS

October 2020 - August 2024

Guest Services Manager

Wilmington, NC

- Managed a 40-person operation and safeguarded \$60,000 - \$80,000 in daily transactions while earning a 95% audit score and consistently a 100% secret-shopper score.
- Responsible for training, team-building, scheduling, and being able to assist any and every guest.